

Notes: Chaney (AER, 2008)

Distorted Gravity: The Intensive and Extensive Margins of International Trade

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1 Big picture

- **Question.** How do variable and fixed trade barriers affect aggregate bilateral trade when firms are heterogeneous and only sufficiently productive firms export?
- **Benchmark overturned.** In representative-firm gravity models, a high elasticity of substitution magnifies the effect of trade barriers. Chaney shows that with Pareto firm heterogeneity, the aggregate effect works differently.
- **Core mechanism.** Lower trade barriers affect trade through two margins: existing exporters sell more, and new firms enter export markets. The first is the intensive margin; the second is the extensive margin.
- **Main result.** A higher elasticity of substitution makes incumbent exporters' sales more sensitive to trade costs, but makes new entrants smaller and therefore weakens the extensive-margin response. Under Pareto productivity, the extensive-margin force dominates.
- **Distorted gravity.** Aggregate bilateral exports retain a gravity structure, but the exponent on variable trade costs is governed by the Pareto shape parameter, not by the elasticity of substitution.
- **Fixed costs.** Fixed export costs enter the gravity equation because they affect the productivity cutoff for exporting. Their effect is inversely related to the elasticity of substitution.
- **Quantitative implication.** Trade barriers inferred from aggregate trade data can be much smaller than representative-firm models imply, because firm heterogeneity amplifies the trade response through entry and exit.
- **Economic takeaway.** The elasticity of substitution is not enough to discipline gravity elasticities. Firm heterogeneity and the extensive margin are central to interpreting trade-cost elasticities, border effects, and sectoral differences in trade flows.

2 Incremental contribution of the paper

- **Incremental contribution.** The paper gives a closed-form many-country gravity model with heterogeneous firms, asymmetric countries, asymmetric bilateral trade barriers, and fixed costs of entering export markets.
- It connects the Melitz selection mechanism to tractable aggregate gravity. Melitz (2003) explains why only productive firms export; Chaney derives how this selection reshapes bilateral trade elasticities in a multi-country gravity environment.
- It shows that the elasticity of substitution has opposite effects on the two margins of trade. Higher substitutability raises the intensive-margin response, but reduces the size of marginal entrants and weakens the extensive-margin response.
- It proves that, under Pareto productivity, the extensive-margin effect dominates. This overturns the representative-firm prediction that higher substitution elasticities mechanically imply larger aggregate trade responses to trade barriers.
- It separates the roles of variable and fixed trade costs analytically. Variable costs affect both intensive and extensive margins; fixed costs affect only entry and hence the extensive margin.
- It offers a reinterpretation of border and distance effects. Large observed trade elasticities need not imply implausibly large trade costs; they can reflect firm selection and the mass of marginal exporters.
- It produces empirically testable sectoral predictions: sectors with more dispersed productivity distributions should show different gravity elasticities and different intensive/extensive margin decompositions than homogeneous sectors.

3 Model primitives and measurement

3.1 Countries, sectors, and preferences

There are N potentially asymmetric countries. Country n has population L_n and labor productivity w_n in a freely traded homogeneous good. The homogeneous good is the numeraire. There are also H differentiated sectors. Consumers have CES preferences within each differentiated sector:

$$U = q_0^{\mu_0} \prod_{h=1}^H \left(\int_{\Omega_h} q_h(\omega)^{(\sigma_h-1)/\sigma_h} d\omega \right)^{\mu_h \sigma_h / (\sigma_h-1)}, \quad \mu_0 + \sum_{h=1}^H \mu_h = 1.$$

Interpretation. The elasticity σ_h governs how easily consumers substitute across varieties inside a sector. The paper's central comparative static asks how this elasticity changes the response of aggregate trade to trade barriers.

3.2 Variable and fixed trade barriers

A firm from country i that sells differentiated-sector good h in country j faces two trade costs:

- an iceberg variable cost $\tau_{ij}^h > 1$, so only $1/\tau_{ij}^h$ units arrive;
- a fixed export cost f_{ij}^h measured in units of the numeraire.

If a firm has productivity φ , the cost of producing q units and delivering them to j is:

$$c_{ij}^h(q) = \frac{w_i \tau_{ij}^h}{\varphi} q + f_{ij}^h.$$

Interpretation. Variable costs reduce the profitability of every unit sold. Fixed costs create selection because only productive firms can cover them.

3.3 Firm productivity distribution

Firms draw productivity from a Pareto distribution:

$$G_h(\varphi) = 1 - \varphi^{-\gamma_h}, \quad \gamma_h > \sigma_h - 1.$$

The parameter γ_h is an inverse measure of heterogeneity. A smaller γ_h means a fatter right tail and more dispersion in productivity.

Interpretation. The Pareto assumption is essential for closed-form distorted gravity. It ensures that the distribution of entrants above a threshold can be summarized by a power law.

3.4 Potential entrants and profits

The mass of potential entrants in country i is proportional to $w_i L_i$. Larger and more productive countries therefore have more potential varieties. The model does not impose free entry. Profits are collected by a global mutual fund and redistributed to workers.

Interpretation. This assumption keeps the model analytically tractable. The number of potential firms is exogenous, while the number of actual exporters is determined by fixed export costs and productivity cutoffs.

3.5 Country remoteness and multilateral resistance

The equilibrium price index in destination j depends on a remoteness term θ_j , which aggregates how difficult it is for all countries to serve market j . It resembles Anderson–van Wincoop multilateral resistance, but it also incorporates firm heterogeneity and fixed entry costs.

Interpretation. A firm’s ability to export to j depends not only on its bilateral cost to j , but also on the toughness of competition in j from all other origins.

4 Models and theoretical specifications

4.1 Benchmark representative-firm gravity

The representative-firm benchmark implies:

$$Exports_{AB} = Constant \times \frac{GDP_A \times GDP_B}{(Trade\ barriers_{AB})^\sigma}.$$

Interpretation. In this benchmark, every firm exports everywhere and trade barriers affect only the quantity sold by each firm. A larger elasticity of substitution makes consumers more sensitive to cost differences, so it magnifies the trade-cost elasticity.

4.2 Firm-level export demand

Given CES demand and monopolistic competition, a firm with productivity φ charges a constant markup over marginal cost. Its exports from i to j are increasing in destination demand and productivity, and decreasing in the wage-adjusted variable trade cost:

$$x_{ij}(\varphi) = \lambda_3 \left(\frac{Y_j}{Y} \right)^{(\sigma-1)/\gamma} \left(\frac{\theta_j}{w_i \tau_{ij}} \right)^{\sigma-1} \varphi^{\sigma-1}, \quad \varphi \geq \bar{\varphi}_{ij}.$$

Interpretation. Conditional on exporting, a firm-level sale looks like the representative-firm model: the elasticity with respect to variable costs is $\sigma - 1$. The aggregate result differs because the set of exporting firms changes.

4.3 Productivity cutoff for exporting

A firm exports only if variable profits cover the fixed cost. The threshold productivity satisfies:

$$\bar{\varphi}_{ij} = \lambda_4 \left(\frac{Y}{Y_j} \right)^{1/\gamma} \left(\frac{w_i \tau_{ij}}{\theta_j} \right) f_{ij}^{1/(\sigma-1)}.$$

Interpretation. Higher variable costs, higher wages, or higher fixed costs raise the export cutoff. A larger destination market lowers the cutoff because it offers more sales over which to spread the fixed cost.

4.4 Aggregate distorted gravity: Proposition 1

Aggregate sector- h exports from country i to country j are:

$$X_{ij}^h = \mu_h \frac{Y_i Y_j}{Y} \left(\frac{w_i \tau_{ij}^h}{\theta_j^h} \right)^{-\gamma_h} (f_{ij}^h)^{-[\gamma_h/(\sigma_h-1)-1]}.$$

Interpretation. This is the paper’s central distorted gravity equation. The exponent on variable trade costs is γ_h , not $\sigma_h - 1$. Aggregate trade is shaped by productivity dispersion, because lower trade costs bring new firms into exporting.

4.5 Intensive and extensive margins

Aggregate exports can be decomposed into:

$$\underbrace{\text{change in exports by existing exporters}}_{\text{intensive margin}} + \underbrace{\text{exports by entering or exiting exporters}}_{\text{extensive margin}}.$$

For a change in variable costs:

$$\zeta_\tau \equiv -\frac{d \ln X_{ij}}{d \ln \tau_{ij}} = \underbrace{(\sigma - 1)}_{\text{intensive}} + \underbrace{[\gamma - (\sigma - 1)]}_{\text{extensive}} = \gamma.$$

For a change in fixed costs:

$$\xi_f \equiv -\frac{d \ln X_{ij}}{d \ln f_{ij}} = \underbrace{0}_{\text{intensive}} + \underbrace{\left[\frac{\gamma}{\sigma - 1} - 1 \right]}_{\text{extensive}}.$$

Interpretation. Variable costs affect both quantities and entry. Fixed costs affect only entry. The elasticity of substitution increases the intensive-margin response, but shrinks the extensive-margin response.

4.6 Proposition 2: the substitution elasticity result

Proposition 2 states:

$$\frac{\partial \zeta_\tau}{\partial \sigma} = 0, \quad \frac{\partial \xi_f}{\partial \sigma} < 0.$$

Interpretation. Contrary to the representative-firm model, a larger elasticity of substitution does not increase the aggregate elasticity with respect to variable costs. For fixed costs, it reduces the aggregate elasticity because marginal entrants become too small to matter much.

5 Theoretical design and identification-style logic

5.1 What the theory is designed to isolate

The paper is theoretical, not an empirical causal design. Its identification-style logic comes from comparing mechanisms inside a controlled model. The model isolates exactly what changes when one moves from representative firms to heterogeneous firms with fixed export costs:

- representative-firm models have only intensive-margin adjustment;
- heterogeneous-firm models add a productivity cutoff and extensive-margin adjustment;
- Pareto productivity makes the size of entrants analytically tractable;
- asymmetric countries and trade barriers preserve the gravity structure while changing its exponents.

Interpretation. The key comparison is between the same CES demand system with and without firm selection. The change in prediction is therefore attributable to firm heterogeneity and fixed export costs.

5.2 Why Pareto productivity matters

The Pareto distribution implies that the mass and size of firms above a threshold have power-law forms. This allows aggregate exports to be solved by integrating firm-level exports over productivity:

$$X_{ij} = w_i L_i \int_{\bar{\varphi}_{ij}}^{\infty} x_{ij}(\varphi) dG(\varphi).$$

Interpretation. The Pareto tail is not merely a mathematical trick. It captures the empirical fact that firm size distributions are highly skewed, so a small number of highly productive firms accounts for much of trade.

5.3 Why the model overturns representative-firm intuition

In a representative-firm model, a high σ makes trade very sensitive to variable costs because varieties are close substitutes. In Chaney's heterogeneous-firm model, high σ also makes productivity differences translate into large size differences. Marginal new exporters are then tiny relative to incumbents, so entry contributes little to aggregate trade.

Interpretation. The representative-firm logic is incomplete because it misses that the identity and size of the marginal exporter depend on the same elasticity of substitution.

5.4 Empirical predictions implied by the model

The model generates several testable implications:

- aggregate trade elasticities should be related to the dispersion of firm productivity;
- the elasticity of firm-level exports with respect to distance or variable costs should depend on σ ;
- the elasticity of the number of exporters should move in the opposite direction;
- sectors with more differentiated goods should have a larger extensive-margin contribution;
- fixed export costs should matter through exporter entry, not through the sales of incumbent exporters.

Interpretation. These predictions connect the paper to later firm-level empirical work that decomposes trade responses into intensive and extensive margins.

5.5 Limitations of the theoretical design

- The model is static and abstracts from dynamic entry, sunk costs, learning, and persistence in export status.
- Productivity is Pareto, which is empirically plausible in the upper tail but still restrictive.
- Fixed costs are exogenous and not micro-founded by marketing, distribution, or information frictions.
- The number of potential entrants is exogenous; the model does not impose free entry.
- The intensive/extensive decomposition is clearest under approximations that ignore the feedback of bilateral trade costs on global price indices for small countries.

Interpretation. These limitations explain why the paper is best read as a benchmark mechanism and gravity derivation rather than as a complete quantitative trade model.

6 Result excerpts

6.1 Original benchmark and main contrast: Krugman gravity vs. distorted gravity

Read: The left excerpt reports the representative-firm gravity benchmark: trade falls with trade barriers raised to an elasticity governed by substitution. The right excerpt reports Chaney's central contrast: once heterogeneous firms and fixed export costs are introduced, the aggregate trade-cost elasticity is negatively related to the substitution elasticity.

Economic message. The comparison shows the paper’s core reversal. In representative-firm models, high substitutability magnifies trade frictions. In Chaney’s model, high substitutability makes marginal entrants small, weakening the extensive margin.

$$Exports_{AB} = Constant \times \frac{GDP_A \times GDP_B}{(Trade\ barriers_{AB})^\sigma},$$

(a) *Original benchmark equation: representative-firm gravity. Read it as the Krugman-style prediction that the substitution elasticity magnifies the effect of trade barriers.*

$$Exports_{AB} = Constant \times \frac{GDP_A \times GDP_B}{(Trade\ barriers_{AB})^{\varepsilon(\sigma)}} \text{ with } \varepsilon'(\sigma) < 0.$$

(b) *Original main contrast: distorted gravity. Read it as the paper’s headline result that the aggregate trade-cost elasticity $\varepsilon(\sigma)$ falls with σ .*

6.2 Original model primitives and equilibrium threshold system

Read: The left excerpt shows the production cost with variable and fixed export costs and the Pareto productivity distribution. The right excerpt shows the equilibrium price index, firm-level exports, export cutoff, country output, and dividends.

Economic message. The model moves from primitives to aggregate gravity by making the export decision a threshold rule. The Pareto distribution then turns the threshold rule into a closed-form aggregate export equation.

$$(8) \quad P_j = \lambda_2 \times Y_j^{1/\gamma - 1/(\sigma - 1)} \times \theta_j,$$

where $\theta_j^{-\gamma} = \sum_{k=1}^N (Y_k/Y) \times (w_k \tau_{kj})^{-\gamma} \times f_{kj}^{-[\gamma/(\sigma-1)-1]}$, Y is world output, and λ_2 is a constant.⁹

An aggregate index of j ’s remoteness from the rest of the world is given by θ_j .¹⁰ It is reminiscent of the “multilateral resistance variable” introduced by Anderson and Eric Van Winco (2003). In addition to their measure, it takes into account the impact of fixed costs and of firm heterogeneity on aggregate prices.

Equilibrium Exports, Thresholds, and Profits.—Exports by an individual firm depend on productivity, the trade barriers it must overcome, aggregate demand, the set of competitors it facing, and the price they set. Plugging the general equilibrium price index from equation (8) in the demand function, and into the productivity threshold from equation (7), I can solve simultaneously for firm level exports the productivity thresholds and total world profits. In general equilibrium, exports $x_{ij}(\varphi)$ from country i to country j by an individual firm with productivity φ , the productivity threshold $\bar{\varphi}_{ij}$ above which firms in i export to j , aggregate output Y_j , and dividend per share π , are given by

$$(9) \quad \left\{ \begin{array}{l} x_{ij}(\varphi) = \begin{cases} \lambda_3 \times \left(\frac{Y_j}{Y}\right)^{(\sigma-1)/\gamma} \times \left(\frac{\theta_j}{w_j \tau_{ij}}\right)^{\sigma-1} \times \varphi^{\sigma-1}, & \text{if } \varphi \geq \bar{\varphi}_{ij} \\ 0 & \text{otherwise,} \end{cases} \\ \bar{\varphi}_{ij} = \lambda_4 \times \left(\frac{Y_j}{Y}\right)^{1/\gamma} \times \left(\frac{w_j \tau_{ij}}{\theta_j}\right) \times f_{ij}^{1/(\sigma-1)}, \end{array} \right.$$

(b) *Original equilibrium system: price index, firm-level exports, and productivity cutoff. This is the machinery that creates extensive-margin selection.*

selling them in country j for a firm with productivity φ is

$$(2) \quad c_{ij}^h(\varphi) = \frac{w_j \tau_{ij}^h}{\varphi} q + f_{ij}^h.$$

Firms are price setters. Given that demand functions are isoelastic, the optimal price charged country j by firm φ from country i is a constant mark-up over the unit cost (including transportation costs): $p_{ij}^h(\varphi) = \sigma_h/(\sigma_h - 1) \times w_j \tau_{ij}^h/\varphi$.

As in Helpman, Melitz, and Yeaple (2004), I assume that productivity shocks are drawn from Pareto distribution with shape parameter γ_h : productivity is distributed over $[1, +\infty)$ according

$$(3) \quad P(\bar{\varphi}_{ij} < \varphi) = G_h(\varphi) = 1 - \varphi^{-\gamma_h},$$

(a) *Original primitives: variable cost, fixed cost, and Pareto productivity. The key objects are τ_{ij} , f_{ij} , and γ .*

6.3 Original Proposition 1: Aggregate Trade and Proposition 2: Intensive and Extensive Margins

Read: Proposition 1 gives the distorted gravity equation. Proposition 2 states that the substitution elasticity has no effect on the variable-cost elasticity of aggregate trade and a negative effect on the fixed-cost elasticity.

Economic message. These two propositions are the main deliverables of the paper. Proposition 1 preserves gravity but changes its exponents. Proposition 2 explains why the exponents differ from representative-firm gravity: intensive and extensive margins move in opposite directions.

PROPOSITION 1 (Aggregate Trade): Total exports (f.o.b.) X_{ij}^h in sector h from country i to country j are given by

$$(10) \quad X_{ij}^h = \mu_h \times \frac{Y_i \times Y_j}{Y} \times \left(\frac{w_i \tau_{ij}^h}{\theta_j^h} \right)^{-\gamma_h} \times (f_{ij}^h)^{-(\gamma_h(\sigma_h-1)-1)}.$$

Exports are a function of country sizes (Y_i and Y_j),¹³ workers' productivity (w_i), the bilateral trade costs, variable (τ_{ij}^h) and fixed (f_{ij}^h), and the measure of j 's remoteness from the rest of the world (θ_j^h).¹⁴

(a) *Original Proposition 1: Aggregate Trade. Variable trade costs enter with exponent $-\gamma_h$, while fixed costs enter through the entry cutoff.*

PROPOSITION 2 (Intensive and Extensive Margins): The elasticity of substitution (σ) has *n* effect on the elasticity of trade flows with respect to variable trade costs (ξ), and a negative effect on the elasticity of trade flows with respect to fixed costs (ζ):

$$\text{if } \zeta = -\frac{d \ln X_{ij}}{d \ln \tau_{ij}} \text{ and } \xi = -\frac{d \ln X_{ij}}{d \ln f_{ij}}, \text{ then } \frac{\partial \zeta}{\partial \sigma} = 0 \text{ and } \frac{\partial \xi}{\partial \sigma} < 0.$$

(b) *Original Proposition 2: Intensive and Extensive Margins. The substitution elasticity does not raise the aggregate variable-cost elasticity and lowers the fixed-cost elasticity.*

6.4 Original margin decompositions: variable costs and fixed costs

Read: The left excerpt decomposes changes in aggregate trade into intensive and extensive components under variable-cost changes. The right excerpt gives the analogous decomposition for fixed costs.

Economic message. Variable costs affect incumbent exporter volumes and firm entry. Fixed costs only move entry. The key result is that the extensive-margin term offsets or dominates the conventional intensive-margin logic.

$$dX_{ij} = \underbrace{\left(w_i L_i \int_{\bar{\varphi}_{ij}}^{\infty} \frac{\partial x_{ij}(\varphi)}{\partial \tau_{ij}} dG(\varphi) \right) d\tau_{ij}}_{\text{Intensive margin}} - \underbrace{\left(w_i L_i x(\bar{\varphi}_{ij}) G'(\bar{\varphi}_{ij}) \times \frac{\partial \bar{\varphi}_{ij}}{\partial \tau_{ij}} \right) d\tau_{ij}}_{\text{Extensive margin}} + \underbrace{\left(w_i L_i \int_{\bar{\varphi}_{ij}}^{\infty} \frac{\partial x_{ij}(\varphi)}{\partial f_{ij}} dG(\varphi) \right) df_{ij}}_{\text{Intensive margin}} - \underbrace{\left(w_i L_i x(\bar{\varphi}_{ij}) G'(\bar{\varphi}_{ij}) \times \frac{\partial \bar{\varphi}_{ij}}{\partial f_{ij}} \right) df_{ij}}_{\text{Extensive margin}}.$$

Following a reduction of trade barriers, each existing exporter ($\varphi > \bar{\varphi}_{ij}$) exports more. This is the intensive margin. At the same time, higher potential profits attract new entrants ($\bar{\varphi}_{ij}$ goes down). This is the extensive margin.

In elasticity notations, I get the following expression for each margin for changes in the variable cost, τ_{ij} .¹⁹

$$\zeta = -\frac{d \ln X_{ij}}{d \ln \tau_{ij}} = (\sigma - 1) + (\gamma - (\sigma - 1)) = \gamma.$$

(a) *Original decomposition for variable costs. The intensive margin is $\sigma - 1$; the extensive margin is $\gamma - (\sigma - 1)$; the total is γ .*

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In elasticity notation, I get the following expression for each margin for changes in the fixed costs, f_{ij} :

$$\xi = -\frac{d \ln X_{ij}}{d \ln f_{ij}} = \underbrace{0}_{\text{Intensive margin Elasticity}} + \underbrace{\frac{\gamma}{\sigma - 1} - 1}_{\text{Extensive margin Elasticity}} = \frac{\gamma}{\sigma - 1} - 1.$$

When only fixed costs move, σ has no impact on the intensive margin, whereas it dampens its impact on the extensive margin ($\gamma/(\sigma - 1) - 1$ decreases with σ). The impact of σ on the elasticity of trade flows with respect to fixed costs is always negative:

(b) *Original decomposition for fixed costs. The intensive margin is zero; the extensive margin is $\gamma/(\sigma - 1) - 1$.*

7 Short summary

- The paper studies how trade barriers affect aggregate trade when firms are heterogeneous and exporting requires both variable and fixed costs.
- The representative-firm benchmark predicts that higher substitution elasticities magnify the impact of trade barriers.
- Chaney shows that this prediction is overturned under Pareto firm heterogeneity.
- Existing exporters respond more strongly to variable costs when σ is high, but new entrants are smaller and matter less.
- The extensive-margin force dominates the intensive-margin force under Pareto productivity.
- Aggregate bilateral trade retains a gravity form, but its exponents are distorted by firm selection.
- Variable trade costs enter with an exponent governed by the Pareto shape parameter γ , not by $\sigma - 1$.
- Fixed export costs matter through the productivity cutoff and the mass of exporters.

- The model implies that aggregate trade can be very responsive to trade costs even if firm-level trade elasticities are moderate.
- It also implies that observed gravity elasticities can be used to learn about firm heterogeneity, not only consumer substitution.

8 Conclusion

8.1 Core conclusion

Chaney shows that firm heterogeneity fundamentally changes the interpretation of gravity equations. Adding fixed export costs and Pareto productivity preserves the gravity structure of bilateral trade flows, but it changes the trade-cost elasticities. Aggregate trade is shaped by the interaction of incumbent exporter responses and the entry or exit of marginal exporters.

8.2 Economic intuition

The intuition is that a fall in trade barriers does two things. Existing exporters sell more, and less productive firms begin exporting. When goods are close substitutes, the intensive effect is strong, but marginal exporters are very small because low productivity is a severe disadvantage. When goods are differentiated, the intensive effect is weaker, but entrants are larger and contribute more to aggregate trade. Under a Pareto firm-size distribution, the entrant-size effect dominates.

8.3 Incremental contribution in the literature

The paper bridges the gap between the Melitz model's firm-level selection mechanism and the gravity literature's aggregate bilateral trade equations. It shows how to embed firm heterogeneity in a multi-country setting with asymmetric barriers while retaining tractability. This is why the paper became a benchmark for structural gravity models with extensive margins.

8.4 Implications for empirical work

The model warns against interpreting gravity coefficients only through the elasticity of substitution. A large distance or tariff coefficient may reflect the mass of marginal exporters rather than high substitutability across goods. Empirical decompositions of trade growth into intensive and extensive margins are therefore essential for understanding trade-policy effects.

8.5 Policy takeaway

Trade liberalization affects both the scale of existing exporters and the set of firms that participate in foreign markets. Policies that reduce fixed entry barriers may have especially large effects on the extensive margin, while policies that reduce variable costs affect both quantities and entry. The welfare and distributional consequences depend on which firms enter, how productive they are, and how large their market shares become.

8.6 Limitations and open questions

The model abstracts from export dynamics, sunk costs, multi-product firms, firm learning, global value chains, and non-Pareto deviations in the productivity distribution. Later work extends these dimensions. Still, the central insight remains: aggregate trade elasticities are not primitive demand elasticities; they are equilibrium objects shaped by firm selection.